

Measurement-Based Multipath Characterization of GPS L1 C/A Signals for Vehicular Satellite-to-Ground Links in Challenging Environments

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Abstract—Vehicular satellite positioning is challenged by multipath and rapidly changing satellite-to-receiver geometry along dense urban and mountain/valley roads. As terrestrial and non-terrestrial systems become increasingly integrated for 6G localization, measurement-based characterization of mobile satellite-to-ground propagation remains important. This paper extracts and statistically characterizes GPS L1 C/A multipath components (MPCs) from moving-vehicle measurements. Navigation-data alignment forms coherent 40-ms observations, SAGE estimates delay, Doppler, and complex gain, and temporal-consistency screening retains persistent MPCs. The analysis uses 518 observations from 236 satellite tracks. Environment-specific models describe excess delay, signed relative Doppler, relative power, and the power-delay relation, with complete-scene complexity checks supporting the reported Doppler and power representations. These path-level descriptions provide inputs for simplified vehicular satellite-to-ground simulation and are relevant to GNSS and NTN-assisted localization under challenging propagation conditions.

Index Terms—GNSS multipath, vehicular satellite-to-ground propagation, GPS L1 C/A, SAGE, NTN-assisted localization, channel modeling.

I. INTRODUCTION

Integrated terrestrial networks (TNs) and non-terrestrial networks (NTNs) are increasingly important to 6G connectivity and localization, while future V2X systems are expected to combine terrestrial and satellite resources [1], [2], [3], [4], [5]. For road vehicles, dense urban streets and mountain/valley terrain create challenging satellite-to-ground propagation: reflections produce delayed replicas, while vehicle and satellite motion introduce Doppler variations that can complicate signal tracking and positioning.

Earlier GNSS studies described multipath through receiver-level quantities such as pseudorange errors and detection indicators [6], [7], while land-mobile satellite models represent time-varying propagation through channel impulse responses or power-delay profiles [8]. At the path level, SAGE estimates individual MPC parameters from complex baseband

measurements [9], [10], and Gaussian mixtures can represent multimodal measured distributions [11]. Path-resolved measurements from moving vehicles therefore complement system-level vehicular and TN/NTN positioning studies by providing empirical delay, Doppler, and power characteristics.

This work applies navigation-data alignment, per-satellite SAGE estimation, and temporal-consistency screening to real moving-vehicle GPS L1 C/A measurements, followed by environment-specific statistical characterization and complete-scene model-complexity checks. Urban and Mountain/Valley measurements are compared without ranking one environment as universally more severe. Although the measurements use GPS L1 C/A rather than an NR-NTN air interface, the resulting path-level evidence is relevant to challenging vehicular satellite-to-ground propagation and NTN-assisted localization evaluation.

II. MEASUREMENTS AND SIGNAL PROCESSING

A. Measurement Setup

The measurement system uses a TEST-TREE RF-Catcher V2 and a roof-mounted GNSS dome antenna. The antenna is right-hand circularly polarized and has a 40-dB active gain. The receiver records GPS L1 C/A at 1575.42 MHz [12]. The sampling rate is 10.23 MHz. Samples are stored as interleaved, little-endian, signed 16-bit I/Q values.

The measurements were collected while the vehicle traveled along two types of roads. Urban routes are surrounded by dense buildings, whereas Mountain/Valley routes pass through mountainous or valley terrain. These route classes are treated as two challenging vehicular satellite-to-ground propagation settings with distinct surrounding structures; the analysis compares their path-level statistics without ranking one environment as universally more severe.

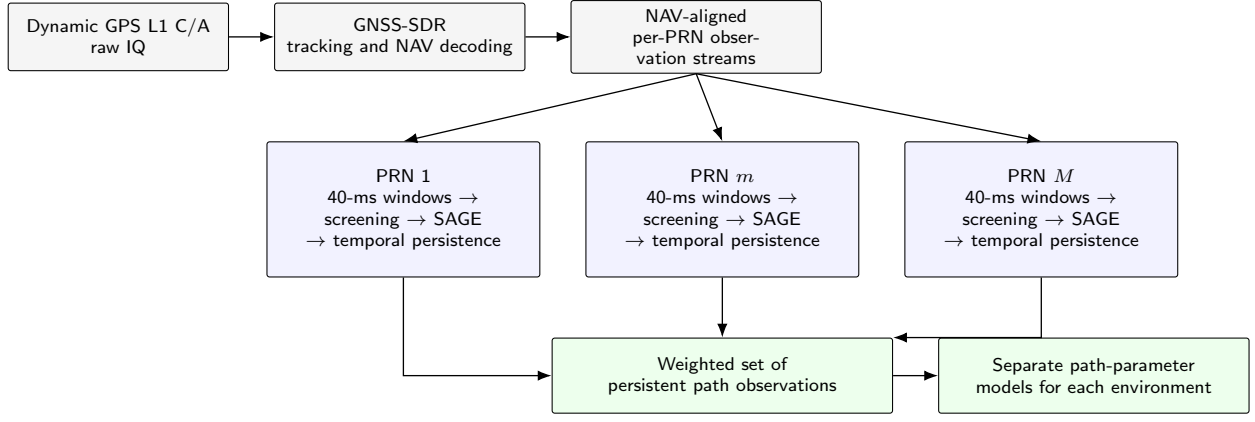


Fig. 1. Measurement and per-satellite MPC processing flow.

B. Per-Satellite Multipath Processing

GNSS-SDR first tracks each visible satellite and decodes its navigation data [13]. Because different satellites use different pseudorandom-noise codes, each tracked satellite is processed in a separate signal stream.

Fig. 1 summarizes this processing flow. Each GPS L1 C/A navigation bit lasts 20 ms; thus, a 40-ms observation may contain a bit transition that reverses the signal sign. The decoded bit signs are removed before coherent observations are formed. Correlation screening identifies observations that may contain an additional signal component. SAGE then estimates MPC parameters for each selected observation. A temporal-consistency check compares adjacent observations and retains persistent MPC observations for the statistical analysis.

III. MULTIPATH COMPONENT ESTIMATION AND STATISTICAL MODELING

A. SAGE-Based MPC Estimation

Let m denote one tracked satellite-to-receiver channel. Its complex baseband signal is

$$r_m(t) = \sum_{\ell=0}^{L_m-1} \alpha_{m,\ell} s_m(t - \tau_{m,\ell}) e^{j2\pi\nu_{m,\ell}t} + n_m(t), \quad (1)$$

where $s_m(t)$ is the aligned C/A-code waveform and $n_m(t)$ is noise. MPC ℓ has complex gain $\alpha_{m,\ell}$, delay $\tau_{m,\ell}$, and Doppler frequency shift $\nu_{m,\ell}$. The earliest estimated component ($\ell = 0$) is used only as a relative reference and is not interpreted as a verified line-of-sight component.

The SAGE algorithm updates one MPC at a time [9], [10]. For target MPC ℓ , the E-step subtracts the current contributions of the other MPCs to form

$$\mathbf{r}_{m,\ell} = \mathbf{r}_m - \sum_{k \neq \ell} \hat{\alpha}_{m,k} \mathbf{q}_m(\hat{\tau}_{m,k}, \hat{\nu}_{m,k}). \quad (2)$$

and the M-step updates its delay and Doppler by

$$(\hat{\tau}_{m,\ell}, \hat{\nu}_{m,\ell}) = \arg \max_{\tau, \nu} \frac{|\mathbf{q}_m(\tau, \nu)^H \mathbf{r}_{m,\ell}|^2}{\mathbf{q}_m(\tau, \nu)^H \mathbf{q}_m(\tau, \nu)}, \quad (3)$$

where $\mathbf{q}_m(\tau, \nu)$ is the sampled delayed and Doppler-shifted code replica. After each sweep, the complex gains are jointly estimated by least squares.

A coarse correlation search initializes the parameters. Delay is refined with replicas spaced by 0.1 sample within ± 0.8 sample, and Doppler is searched within ± 30 Hz of the tracking estimate in 5-Hz steps. Iterations stop after 10 sweeps or when the relative RSS change falls below 10^{-6} . For each 40-ms observation, candidate path counts $L = 1-4$ are compared using BIC and RSS.

B. Persistent MPC Observations

Adjacent observations are matched when their excess-delay, signed relative Doppler, and relative-power differences are no greater than 1.5 samples, 40 Hz, and 10 dB, respectively. An MPC is persistent when matches occur in at least three consecutive observations; a 40-ms observation enters the modeling set only when every nonreference MPC satisfies this criterion. The center-time parameters are retained as the persistent MPC observation.

For each persistent MPC observation, the three parameters are expressed relative to the reference component:

$$\Delta\tau = \tau_\ell - \tau_0, \quad \nu_{\text{rel}} = \nu_\ell - \nu_0, \quad P_{\text{rel}} = 10 \log_{10} \frac{|\alpha_\ell|^2}{|\alpha_0|^2}. \quad (4)$$

Here, $\Delta\tau$ is the delay relative to the reference component, ν_{rel} is signed relative Doppler, and P_{rel} is the power relative to the reference component in decibels.

C. Candidate Models and Environment-Specific Statistical Descriptions

Satellite tracks contain different numbers of persistent MPC observations. Each track is therefore assigned the same total weight, preventing a longer track from dominating the environment-level statistics simply because it contains more observations. If track j contains n_j observations, each observation in that track receives weight $w_i = 1/n_j$.

Candidate distributions for excess delay and relative power were compared by leave-one-scene-out (LOSO) validation,

with complete scenes held out in turn. Model selection considered predictive likelihood and the maximum weighted CDF deviation,

$$D_w = \max_i \left| \hat{F}_w(x_i) - F_\theta(x_i) \right|, \quad (5)$$

where $\hat{F}_w$ is the weighted empirical CDF and F_θ is the fitted model CDF; smaller D_w indicates closer descriptive agreement. One-, two-, and three-component GMMs were also evaluated for signed relative Doppler and relative power using the same track weighting and complete-scene validation. The one-standard-error rule selected diagnostic complexities of $K = 2$ and $K = 3$ for Urban and Mountain/Valley Doppler, respectively, and $K = 2$ for relative power in both environments. Doppler mixtures are used only as complexity diagnostics; the reported Doppler representation remains the track-balanced empirical CDF.

Signed relative Doppler is represented directly by its track-balanced empirical CDF. For environment e , the CDF and its weighted quantiles are defined as

$$\hat{F}_{\nu,e}(x) = \frac{\sum_{i \in e} w_i \mathbf{1}(\nu_i \leq x)}{\sum_{i \in e} w_i}, \quad (6)$$

$$P_q(e) = \inf\{x : \hat{F}_{\nu,e}(x) \geq q\}.$$

Here, ν_i is the signed relative Doppler of observation i . The empirical representation retains the measured signed relative Doppler values and is summarized by P_{10} , P_{50} , and P_{90} .

To represent the observed positive and right-skewed excess-delay distribution, a lognormal form is used:

$$\ln(\Delta\tau - \tau_s) \sim \mathcal{N}(\mu_\tau, \sigma_\tau^2), \quad (7)$$

where τ_s is a shift, $\exp(\mu_\tau)$ is the scale, and σ_τ controls the spread in the logarithmic domain. The Urban model uses an estimated shift, while the Mountain/Valley model uses $\tau_s = 0$.

A two-component Gaussian mixture model (GMM) is used to represent the bimodal structure of the measured relative-power distribution:

$$p(x) = \pi \mathcal{N}(x; \mu_1, \sigma_1^2) + (1 - \pi) \mathcal{N}(x; \mu_2, \sigma_2^2). \quad (8)$$

The mixing weight π gives the relative contribution of the first Gaussian component.

Relative power is also modeled as a function of excess delay. Two line segments with different slopes meet at a common breakpoint τ_b :

$$P(\tau) = \begin{cases} P_b + b_1(\tau - \tau_b), & \tau < \tau_b, \\ P_b + b_2(\tau - \tau_b), & \tau \geq \tau_b. \end{cases} \quad (9)$$

This connected model allows the mean rate of power decrease to differ between low-delay and high-delay MPCs while keeping the mean trend continuous. The fitted delay and power distributions and the track-balanced empirical CDF of signed relative Doppler provide complementary parameter descriptions for each environment.

IV. STATISTICAL MODELING RESULTS

A. Modeling Data and Excess Delay

The analysis uses 518 persistent MPC observations from 236 satellite tracks, 36 recordings, and 9 measurement scenes. Of these observations, 346 were collected in the Urban environment and 172 in the Mountain/Valley environment.

Fig. 2(a) and (b) show that both excess-delay distributions are right-skewed. The Urban fit has a shift of 68.7 ns, a scale of 102.8 ns, and a log-domain standard deviation of 0.675. The Mountain/Valley fit has a scale of 187.8 ns and a log-domain standard deviation of 0.348. The larger log-domain standard deviation in Urban is consistent with its broader right tail.

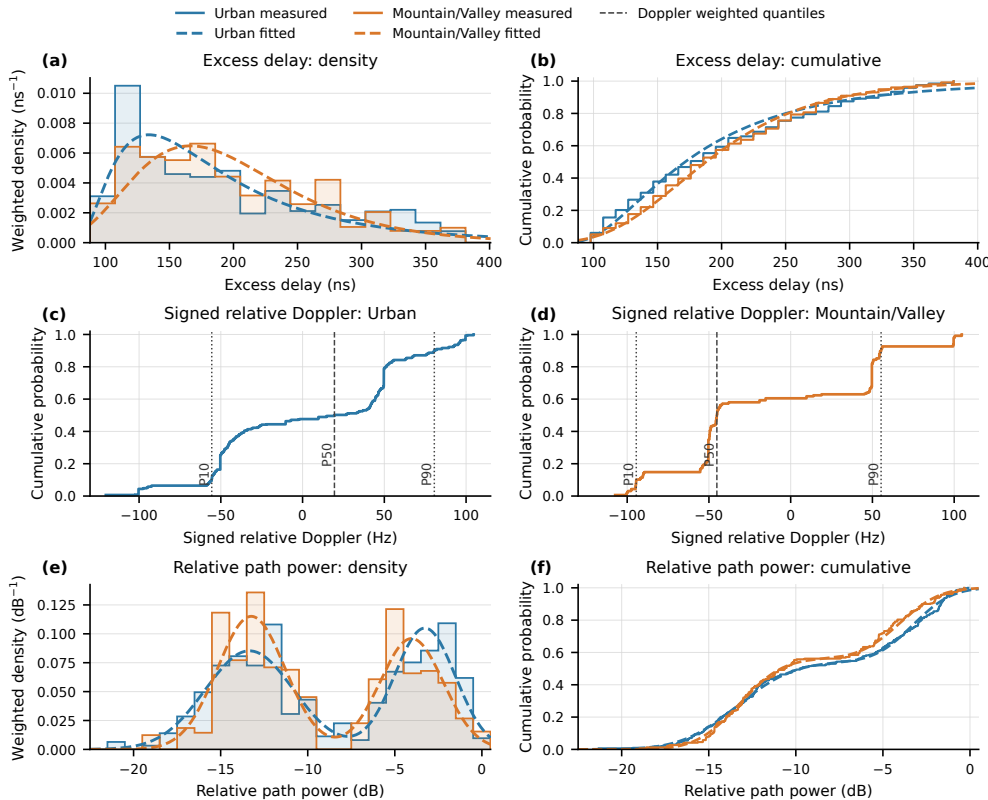


Fig. 2. Measured path-parameter distributions: excess delay (a,b), signed relative Doppler ECDFs (c,d), and relative power (e,f) for Urban and Mountain/Valley.

B. Signed Relative Doppler

Fig. 2(c) and (d) show the track-balanced signed relative Doppler ECDFs. Urban has weighted P_{10} , P_{50} , and P_{90} values of -55.6 , 19.5 , and 80.6 Hz, compared with -94.4 , -45.1 , and 55.3 Hz for Mountain/Valley. Mountain/Valley has a more negative P_{10} and median, whereas the Urban median remains positive. Complete-scene complexity checks selected diagnostic GMM orders $K = 2$ and $K = 3$, respectively, but the empirical CDF remains the reported representation to avoid imposing a parametric mixture law.

The fitted model summaries are collected in Table I.

TABLE I
ENVIRONMENT-SPECIFIC PATH-PARAMETER DESCRIPTIONS.

Quantity	Urban	Mountain/Valley
Excess delay	Shifted lognormal: shift 68.7 ns, scale 102.8 ns, log-std. 0.675, $D_w = 0.082$	Lognormal: scale 187.8 ns, log-std. 0.348, $D_w = 0.055$
Signed relative Doppler	Empirical CDF: $P_{10} = -55.6$, $P_{50} = 19.5$, $P_{90} = 80.6$ Hz; diagnostic $K = 2$	Empirical CDF: $P_{10} = -94.4$, $P_{50} = -45.1$, $P_{90} = 55.3$ Hz; diagnostic $K = 3$
Relative power	2-GMM: $(-13.35, 2.53, 0.541)$, $(-3.27, 1.74, 0.459)$; $D_w = 0.045$	2-GMM: $(-13.24, 1.95, 0.562)$, $(-4.03, 1.82, 0.438)$; $D_w = 0.037$
Power-delay	Breakpoint 190.6 ns; slopes $-3.53/-5.57$ dB/100 ns; RMSE 4.01 dB	Breakpoint 190.6 ns; slopes $-6.31/-4.46$ dB/100 ns; RMSE 3.44 dB

GMM tuples denote (mean, standard deviation, weight). D_w is the maximum weighted CDF deviation. Doppler K values are diagnostic only; the reported Doppler model is the empirical CDF.

D. Power-Delay Relation

Fig. 3 shows the decrease in relative power with increasing excess delay. The connected model has a common breakpoint of 190.6 ns, with separate fitted mean power-decay slopes below and above the breakpoint. In Urban, the slopes are -3.53 and -5.57 dB per 100 ns before and after the breakpoint; in Mountain/Valley, they are -6.31 and -4.46 dB per 100 ns. The fitted mean trend is steeper after the breakpoint in Urban and less steep after it in Mountain/Valley. The root-mean-square errors are 4.01 and 3.44 dB, respectively.

For simplified GNSS channel simulation, excess delay and relative power can be sampled from their fitted marginals, signed relative Doppler can be resampled from its track-balanced empirical CDF, and the connected power-delay model can supply the mean delay-dependent power. Together, these descriptions provide MPC parameters for each simulated satellite channel; they do not specify occurrence rates, temporal evolution, or a complete joint stochastic channel model.

These measurements use GPS L1 C/A rather than an NR-NTN air interface, so their NTN relevance is limited

C. Relative Power

Fig. 2(e) and (f) show two relative-power modes in both environments. The mode means are near -13 dB and -3 to -4 dB, relative to the earliest component. Compared with a single Gaussian, the two-component GMM reduces D_w from 0.133 to 0.045 in Urban and from 0.137 to 0.037 in Mountain/Valley; complete-scene complexity checks also select $K = 2$ in both environments.

to satellite-to-ground propagation and positioning. The path-level statistics can support simplified evaluation of vehicular satellite positioning and integrated TN/NTN localization [3], [4], [5].

V. CONCLUSION

This paper characterizes GPS L1 C/A multipath at the path level using moving-vehicle measurements in challenging Urban and Mountain/Valley environments. Navigation-data alignment, per-satellite SAGE estimation, and temporal-consistency screening yield persistent MPC observations whose excess delay, signed relative Doppler, relative power, and power-delay behavior are described by environment-specific models. Complete-scene complexity checks support retaining the empirical-CDF Doppler representation and the two-component relative-power GMM. The resulting measurements provide inputs for simplified vehicular satellite-to-ground simulation and empirical propagation evidence relevant to NTN-assisted localization without constituting an NR-NTN channel model.

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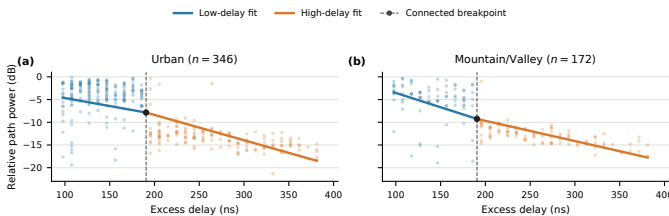


Fig. 3. Measured relative power and connected two-slope fits versus excess delay.

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